SCAFFOLD DESIGN REPORT

NIGERIA LIQUEFIED NATURAL GAS LIMITED

LOCATION: Jetty-1	**AREA: B**
PERMIT TO WORK No.: -	**TYPE OF SCAFFOLD:** Independent Return
SCAFFOLD DRAWING No: PMS-WA-ARCO-038	
RISK CLASSIFICATION: ✓ High ○ Medium ○ Low	

SCAFFOLD TO BE INSPECTED BY PIC BEFORE TAGGING

BRIEF DESCRIPTION OF SCAFFOLD DESIGN

This document describes the structural design of an Independent Return scaffold required for maintenance of the condensate loading arm, equipment tag Z-3301A,Z-3301B,Z-3301C at Jetty-1. The scaffold envelope is 5m x 4m x 20m (length x width x height) and is configured with standards, ledgers, transoms, bracing, platforms, and support/tie arrangements as indicated on the approved drawing.

SAFE WORKING LOAD (SWL): 1.5 kN/m²

Total Horizontal Load in the X Direction (F_X): 2.103 kN
Total Horizontal Load in the Z Direction (F_Z): 4.807 kN

Role	ID	Name	Signature	Date
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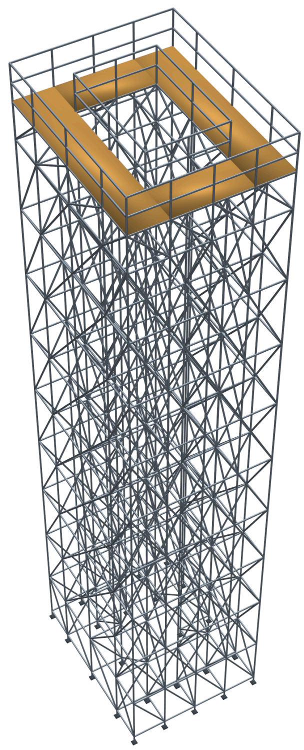
TABLE OF CONTENTS

CALCULATION SHEETS

ENGINEERING DRAWINGS

3D ISOMETRIC VIEW



GENERAL NOTE

BS EN 12811-1, 4.2.1.2 / 10.3.2.2

Scaffold tube material has a minimum nominal yield strength of 235 N/mm² (BS EN 12811-1, 4.2.1.2). A partial safety factor γ_M = 1.1 (BS EN 12811-1, 10.3.2.2) has been applied, giving a design resistance of 213.6 N/mm² used in member capacity checks. This design calculation was carried out in accordance with the limit states design method stipulated in EN 1993-1-1:2005 (Eurocode 3 - Design of Steel Structures).

Material Properties

48.3 mm outside diameter x 3.6 mm thick steel tube was defined by STAAD.Pro Connect Edition V22 section tables (British) and assigned to all scaffold members as section **PIP483.6**.

Modelling Philosophy

BS EN 12811-1, 10.2.3.2 / 10.2.3.3

A 3-D model was analysed using **STAAD.Pro CONNECT Edition (V22)** in compliance with EN 1993-1-1:2005. Base supports are modelled as pinned with lateral spring stiffness representing frictional restraint (KFX/KFZ = 17.522 kN). Base supports are modelled as elastic spring supports, restrained laterally in both horizontal directions with rotational degrees of freedom released, representing flexible ground contact conditions at each standard foot.

LOAD CASES

Dead Load / Self-weight

Self-weight of structural elements computed by STAAD.Pro with steel density **76.8195 kN/m³**, with a **10% allowance** for fittings and couplers (selfweight factor = 1.1).

Live Load on Platforms

BS EN 12811-1

Platform live loading is applied to the scaffold working platforms as member UDLs in the STAAD model. The load intensity is multiplied by the tributary width of each supporting member to obtain the applied member line load. (LC 2).

Member	Tributary Width (m)	Load Intensity (kN/m²)	Member Load (kN/m)
Members 661, 664, 667, 671, 681, 682, 686, 692 (Edge)	1.000 / 2 = 0.500	1.500	**0.750**
Members 669, 684 (Interior)	(0.500 + 0.500) = 1.000	1.500	**1.500**
Members 666, 672 (Interior)	(0.500 + 0.750) = 1.250	1.500	**1.875**
Members 678, 688 (Interior)	(0.750 + 0.500) = 1.250	1.500	**1.875**
Members 675, 690 (Interior)	(0.750 + 0.750) = 1.500	1.500	**2.250**
*Load Intensity = **1.50 kN/m²** — BS EN 12811-1 Class 2*			

Member Load = Load Intensity × Tributary Width. Edge members carry half the adjacent bay width; interior members carry half the bay from each side.
Governing load intensity (SWL) = **1.5 kN/m²** — BS EN 12811-1 Class 2.
Total applied platform live load = **21.0 kN**.

Handrail Load

Handrail loads were applied where required by the scaffold configuration and modelled independently in the global X and Z directions.

Handrail Load (HLX) = **3.6 kN** (member UDL 0.15 kN/m)
Handrail Load (HLZ) = **4.8 kN** (member UDL 0.15 kN/m)

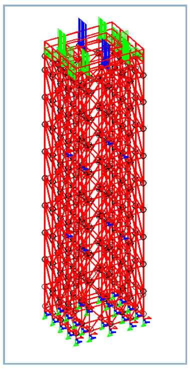
Wind Load

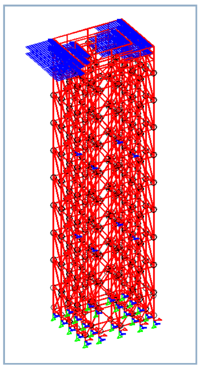
EN 1991-1-4:2005

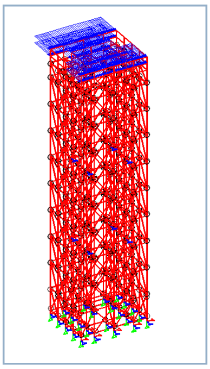
Wind loads were derived in accordance with EN 1991-1-4:2005 using NLNG Bonny Island site parameters. Full calculation chain is given in the Wind Load Calculation sheet below. Wind UDL = **0.077357 kN/m** applied as a member UDL to all exposed scaffold members in the global X and Z wind directions.

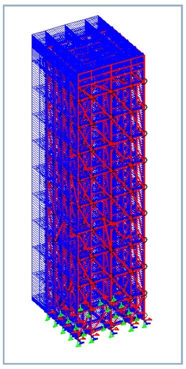
Total Wind Load - X Axis (WLX) = **54.05 kN**
Total Wind Load - Z Axis (WLZ) = **58.06 kN**

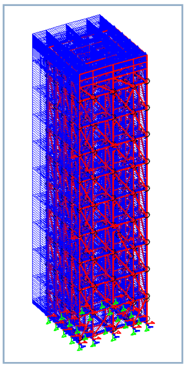
Load Diagrams


Live Load (LL)


Handrail Load X (HLX)


Handrail Load Z (HLZ)


Wind Load X (WLX)


Wind Load Z (WLZ)

WIND LOAD CALCULATION - EN 1991-1-4:2005

Reference	Calculation	Value	Unit
	Basic wind speed (3-second gust)		
	V3 =	**36.0**	m/s
	Eurocode $V_{b,0}$ is the characteristic 10-minute mean wind velocity		
Durst, C.S. (1960)	Convert 3-second gust to 10-minute mean wind velocity		
	V3 / V3600 =	1.52	
	V600 / V3600 =	1.06	
	V600 = (V600/V3600) x (V3600/V3) x V3 =	**25.11**	m/s
	Basic wind speed V_b = V600 =	**25.11**	m/s
EN 1991-1-4:2005	Roughness Length (z0) - Terrain Category 0	0.003	m
	Terrain category II (z0,ii)	0.05	
EN 1991-1-4, Table 4.1	z_{min}	1.0	m

Reference	Calculation	Value	Unit
	z - height of structure under consideration	**20.0**	m
EN 1991-1-4, Sec 4.3	$C_{r0}(z)$ - Roughness factor	1.0	
EN 1991-1-4, Sec 4.3	$C_o(z)$ - Orography factor	1.0	
EN 1991-1-4, Sec 4.3	Terrain factor $K_r = 0.19 \times [z0/z0,ii]^{0.07}$	0.156	
EN 1991-1-4, Table 4.1	$z_{min} < z$ -> Case 1 applies		
	Wind Turbulence		
	$C_r(z) = K_r \times \ln(z/z0)$	1.3739	
	Mean velocity $V_m(z) = C_r(z) \times C_o(z) \times V_b$	34.492	m/s
EN 1991-1-4, Sec 4.4	K1	1.0	
	Turbulence intensity $I_v(z) = k1 / [C_o(z) \times \ln(z/z0)]$	0.1136	
	Peak Velocity Pressure		
	Density of air rho	1.25	kg/m³
EN 1991-1-4, Sec 4.5	$q_p(z) = [1 + 7*I_v(z)] \times 0.5 \times rho \times V_m(z)^2$	**1334.6649**	N/m²
Wind Load = $C_f \times q_p \times A_f$			
C_f =		1.2	
q_p =		1.334665	kN/m²
A_f =	Tube outer diameter (48.3 mm)	0.0483	m
Wind Load UDL =		**0.077357**	**kN/m**

LOAD COMBINATIONS

BS EN 12811-1, 6.2.9.2

Load combinations in accordance with BS EN 12811-1:2003. ULS combinations use gamma = **1.5**; SLS (characteristic) combinations use gamma = **1.0**.

DL = Dead Load | **LL** = Platform Live Load | **HLX/HLZ** = Handrail X/Z | **WLX/WLZ** = Wind X/Z

LC No.	Combination	Component Loadings & Factors
7	**SLS DL + LL**	LC1 x 1.0 + LC2 x 1.0
8	**SLS DL + LL + HLX**	LC1 x 1.0 + LC2 x 1.0 + LC3 x 1.0
9	**SLS DL + LL + HLZ**	LC1 x 1.0 + LC2 x 1.0 + LC4 x 1.0
10	**SLS DL + WLX**	LC1 x 1.0 + LC5 x 1.0
11	**SLS DL + WLZ**	LC1 x 1.0 + LC6 x 1.0
12	**SLS DL + LL + WLX**	LC1 x 1.0 + LC2 x 1.0 + LC5 x 1.0
13	**SLS DL + LL + WLZ**	LC1 x 1.0 + LC2 x 1.0 + LC6 x 1.0
14	**SLS DL + LL + WLZ + HLX**	LC1 x 1.0 + LC2 x 1.0 + LC6 x 1.0 + LC3 x 1.0
15	**SLS DL + LL + WLZ + HLZ**	LC1 x 1.0 + LC2 x 1.0 + LC6 x 1.0 + LC4 x 1.0
16	**ULS 1.5DL + 1.5LL**	LC1 x 1.5 + LC2 x 1.5
17	**ULS 1.5DL + 1.5LL + 1.5HLX**	LC1 x 1.5 + LC2 x 1.5 + LC3 x 1.5
18	**ULS 1.5DL + 1.5LL + 1.5HLZ**	LC1 x 1.5 + LC2 x 1.5 + LC4 x 1.5
19	**ULS 1.5DL + 1.5LL + 1.5WLX**	LC1 x 1.5 + LC2 x 1.5 + LC5 x 1.5
20	**ULS 1.5DL + 1.5LL + 1.5WLZ**	LC1 x 1.5 + LC2 x 1.5 + LC6 x 1.5
21	**ULS 1.5DL + 1.5LL + 1.5WLZ + 1.5HLX**	LC1 x 1.5 + LC2 x 1.5 + LC6 x 1.5 + LC3 x 1.5
22	**ULS 1.5DL + 1.5LL + 1.5WLZ + 1.5HLZ**	LC1 x 1.5 + LC2 x 1.5 + LC6 x 1.5 + LC4 x 1.5
23	**ULS 1.5DL + 1.5WLX**	LC1 x 1.5 + LC5 x 1.5
24	**ULS 1.5DL + 1.5WLZ**	LC1 x 1.5 + LC6 x 1.5
25	**ULS DL + 1.5WLX**	LC1 x 1.0 + LC5 x 1.5

LC No.	Combination	Component Loadings & Factors
26	**ULS DL + 1.5WLZ**	LC1 x 1.0 + LC6 x 1.5

STATIC CHECK PARAMETERS - UTILIZATION RATIO SUMMARY

Maximum Utilization Ratio = 0.352 at Member No. **1500**, Load Combination **22**, Clause **EC-6.2.9.1**. **PASS** - All members adequate; UC ratio below unity.

Member	Section	Status	Clause	UC Ratio	L/C	F_x (kN)	Type
1500	PIP483.6	PASS	EC-6.2.9.1	**0.352**	22	0.26	C
1310	PIP483.6	PASS	EC-6.3.3-661	**0.312**	22	11.97	C
1482	PIP483.6	PASS	EC-6.2.9.1	**0.312**	22	0.19	C
477	PIP483.6	PASS	EC-6.3.3-662	**0.310**	22	13.40	C
690	PIP483.6	PASS	EC-6.2.9.1	**0.304**	22	0.91	T
1304	PIP483.6	PASS	EC-6.3.3-661	**0.304**	22	11.66	C
1497	PIP483.6	PASS	EC-6.2.9.1	**0.301**	22	0.20	C
462	PIP483.6	PASS	EC-6.3.3-662	**0.300**	22	12.91	C
1487	PIP483.6	PASS	EC-6.2.9.1	**0.284**	22	0.27	C
688	PIP483.6	PASS	EC-6.3.3-662	**0.265**	22	0.28	C
1309	PIP483.6	PASS	EC-6.3.3-661	**0.265**	22	10.08	C
672	PIP483.6	PASS	EC-6.3.3-662	**0.255**	22	1.73	C
678	PIP483.6	PASS	EC-6.3.3-662	**0.251**	22	1.80	C
666	PIP483.6	PASS	EC-6.2.9.1	**0.247**	22	0.03	C
675	PIP483.6	PASS	EC-6.2.9.1	**0.247**	22	0.21	T
1490	PIP483.6	PASS	EC-6.2.9.1	**0.245**	22	0.21	C
478	PIP483.6	PASS	EC-6.3.3-662	**0.244**	22	10.04	C
463	PIP483.6	PASS	EC-6.3.3-662	**0.240**	22	9.82	C
647	PIP483.6	PASS	EC-6.3.3-662	**0.240**	22	0.89	C
1486	PIP483.6	PASS	EC-6.2.9.1	**0.240**	22	0.22	C
473	PIP483.6	PASS	EC-6.3.3-662	**0.223**	22	8.91	C
1327	PIP483.6	PASS	EC-6.3.3-661	**0.223**	19	6.65	C
464	PIP483.6	PASS	EC-6.3.3-662	**0.221**	22	8.82	C
1303	PIP483.6	PASS	EC-6.3.3-661	**0.220**	22	8.32	C
1201	PIP483.6	PASS	EC-6.3.3-661	**0.216**	21	8.16	C
1219	PIP483.6	PASS	EC-6.3.3-661	**0.215**	19	6.41	C
442	PIP483.6	PASS	EC-6.2.9.1	**0.207**	22	0.17	T
480	PIP483.6	PASS	EC-6.3.3-662	**0.206**	22	8.84	C
1273	PIP483.6	PASS	EC-6.3.3-661	**0.206**	22	7.77	C
427	PIP483.6	PASS	EC-6.2.9.1	**0.203**	22	0.43	T
431	PIP483.6	PASS	EC-6.3.3-662	**0.200**	22	0.17	C
650	PIP483.6	PASS	EC-6.3.3-662	**0.200**	22	1.28	C
1213	PIP483.6	PASS	EC-6.3.3-661	**0.199**	19	5.86	C
1328	PIP483.6	PASS	EC-6.3.3-661	**0.197**	19	7.41	C
1489	PIP483.6	PASS	EC-6.2.9.1	**0.197**	22	0.25	T
420	PIP483.6	PASS	EC-6.3.3-662	**0.196**	22	8.34	C
1196	PIP483.6	PASS	EC-6.3.3-661	**0.196**	22	7.35	C
1485	PIP483.6	PASS	EC-6.2.9.1	**0.196**	22	0.24	T
452	PIP483.6	PASS	EC-6.2.9.1	**0.195**	22	1.94	T
646	PIP483.6	PASS	EC-6.3.3-662	**0.195**	22	1.10	C

Member	Section	Status	Clause	UC Ratio	L/C	F_x (kN)	Type
1212	PIP483.6	PASS	EC-6.3.3-661	**0.195**	19	5.74	C
416	PIP483.6	PASS	EC-6.3.3-662	**0.192**	22	8.19	C
1195	PIP483.6	PASS	EC-6.3.3-661	**0.190**	21	7.12	C
1194	PIP483.6	PASS	EC-6.3.3-661	**0.189**	20	7.06	C
475	PIP483.6	PASS	EC-6.3.3-662	**0.188**	19	7.66	C
472	PIP483.6	PASS	EC-6.3.3-662	**0.187**	19	7.56	C
1321	PIP483.6	PASS	EC-6.3.3-661	**0.186**	19	5.46	C
1494	PIP483.6	PASS	EC-6.2.9.1	**0.186**	21	0.22	C
297	PIP483.6	PASS	EC-6.3.3-662	**0.185**	22	7.69	C
537	PIP483.6	PASS	EC-6.3.3-662	**0.185**	22	7.57	C
Showing highest 50 utilization ratios; 1021 lower-utilization members omitted.							

SCAFFOLD TUBES CONNECTION STABILITY CHECK

EN 12811, Table C.1

The maximum axial load on scaffold tube members (from STAAD.Pro Connect Edition analysis) is **3.740 kN** (Member 428, C). The required right-angle coupler class is selected from Table C.1 below using the slipping resistance limit.

Selected coupler class: **Class A** (capacity = 10.0 kN).

Check: 3.740 kN <= 10.0 kN -> **PASS**

Class A right-angle couplers are adequate.

Table C.1 - Characteristic Values of Resistances for Couplers (EN 12811)

Coupler Type	Resistance	Characteristic Value	
		Class A	Class B
Right-angle Coupler (RA)	Slipping force $F_{s,k}$ (kN)	**10.0**	**15.0**
	Cruciform bending moment $M_{B,k}$ (kNm)	-	0.8
	Pull-apart force $F_{p,k}$ (kN)	20.0	30.0
	Rotational moment $M_{T,k}$ (kNm)	-	0.13
Swivel Coupler (SW)	Slipping force $F_{s,k}$ (kN)	6.0	9.0
	Cruciform bending moment $M_{B,k}$ (kNm)	-	0.8
	Pull-apart force $F_{p,k}$ (kN)	15.0	22.5
	Rotational moment $M_{T,k}$ (kNm)	-	0.08
Putlog / Single Coupler (PL)	Slipping force $F_{s,k}$ (kN)	-	3.0
	Cruciform bending moment $M_{B,k}$ (kNm)	-	0.6

Member Axial Force Summary (Top 30 Horizontal Members — ULS Code Check)

Member	Type	Status	Clause	UC Ratio	L/C	Axial Force (kN)
428	C	PASS	EC-6.3.3-662	0.117	22	**3.740**
503	C	PASS	EC-6.3.3-662	0.097	22	**3.520**
683	C	PASS	EC-6.3.3-662	0.076	22	**3.250**
1471	C	PASS	EC-6.3.3-661	0.069	22	**3.120**
490	C	PASS	EC-6.3.3-662	0.086	22	**3.080**
563	C	PASS	EC-6.3.3-662	0.061	22	**3.020**
550	C	PASS	EC-6.3.3-662	0.063	22	**2.710**
548	C	PASS	EC-6.3.3-662	0.058	22	**2.680**
668	C	PASS	EC-6.3.3-662	0.118	22	**2.570**

Member	Type	Status	Clause	UC Ratio	L/C	Axial Force (kN)
629	T	PASS	EC-6.2.9.1	0.100	22	**2.410**
608	C	PASS	EC-6.3.3-662	0.056	22	**2.340**
696	C	PASS	EC-6.3.3-662	0.103	22	**2.240**
569	T	PASS	EC-6.2.9.1	0.114	22	**2.230**
573	C	PASS	EC-6.3.3-662	0.058	22	**2.110**
693	C	PASS	EC-6.3.3-662	0.111	22	**2.060**
451	T	PASS	EC-6.2.9.1	0.160	22	**2.030**
633	C	PASS	EC-6.3.3-662	0.065	22	**1.950**
452	T	PASS	EC-6.2.9.1	0.195	22	**1.940**
678	C	PASS	EC-6.3.3-662	0.251	22	**1.800**
509	T	PASS	EC-6.2.9.1	0.134	22	**1.760**
677	C	PASS	EC-6.3.3-662	0.136	22	**1.750**
672	C	PASS	EC-6.3.3-662	0.255	22	**1.730**
512	T	PASS	EC-6.2.9.1	0.125	22	**1.700**
623	C	PASS	EC-6.3.3-661	0.045	22	**1.700**
1474	C	PASS	EC-6.3.3-662	0.044	21	**1.680**
391	T	PASS	EC-6.2.9.1	0.131	22	**1.630**
607	T	PASS	EC-6.2.9.1	0.072	26	**1.610**
425	T	PASS	EC-6.2.9.1	0.168	22	**1.600**
496	T	PASS	EC-6.2.9.1	0.129	22	**1.580**
493	T	PASS	EC-6.2.9.1	0.119	22	**1.530**

FRICTIONAL RESISTANCE

Max vertical reaction per support (Max f_y) = Resistance x 0.15 / C_F	= **8.761 kN**
Coefficient of friction (C_F)	= **0.3**
Frictional Resistance (FR) = C_F x Max f_y	= **2.6283 kN**
Spring Stiffness = FR / 0.15	= **17.522 kN** <- applied as KFX/KFZ in STAAD model

DEFLECTION CHECK RESULTS

BS EN 12811-1, Cl 6.3 Serviceability limit state — values from STAAD.Pro SLS (unfactored) combination results. Displacement tables are shown below.

Vertical Deflection Check

Parameter	Value	Allowable	Check
Max vertical displacement - LC 15 Member 1557 | Member length L = 1500.0 mm	**0.798 mm**	L/100 = 1500/100 = **15.0 mm**	PASS

Vertical Displacement Table (SLS Combinations)

Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
1557	0.790	0.750	15	1897	0.120	0.074	9.322
1560	0.787	0.750	15	1906	0.119	0.074	9.403
675	0.657	0.417	15	1522	0.132	-0.688	4.444
1571	0.630	0.750	15	2382	-0.214	-0.851	7.999
1568	0.608	0.750	15	2468	-0.214	-0.859	7.983
675	0.578	0.417	9	1728	-0.040	-0.856	1.110
675	0.548	0.500	13	1824	0.070	-0.710	3.284
675	0.546	0.500	14	1829	0.389	-0.707	3.332
1509	0.527	0.875	15	2845	0.121	0.078	6.486
1560	0.518	0.750	9	2893	-0.096	-0.248	4.614
1512	0.518	0.875	15	2896	0.120	0.079	6.533
1557	0.514	0.750	9	2918	-0.095	-0.257	4.552
678	0.511	0.583	15	1957	0.175	-0.540	2.721
672	0.504	0.583	15	1982	0.083	-0.529	2.621

Max = **0.798 mm** (LC 15) | L = Member 1557 length (1500 mm) | Allowable L/100 = 15.0 mm → **PASS**

Horizontal Deflection Check

Parameter	Value	Allowable	Check
Max horizontal displacement - LC 15 Member 647 | Member length L = 1000.0 mm	**0.425 mm**	L/200 = 1000/200 = **5.0 mm**	**PASS**

Horizontal Displacement Table (SLS Combinations)

Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
647	0.425	0.667	15	2352	0.125	0.110	3.898
647	0.342	0.667	9	2921	-0.090	-0.217	0.714
650	0.337	0.667	15	2970	0.121	0.090	2.288
646	0.332	0.667	15	3013	0.121	0.090	2.195
587	0.303	1.167	15	6600	0.136	0.118	4.108
590	0.275	1.167	15	7279	0.139	0.107	2.401
586	0.273	1.167	15	7314	0.130	0.104	2.320
1500	0.268	0.208	15	1868	-0.239	-0.828	4.462
647	0.260	0.583	14	3848	0.313	-0.040	2.980
647	0.259	0.583	13	3855	0.029	-0.052	2.939
587	0.246	1.167	13	8142	0.049	-0.036	3.162
587	0.243	1.167	14	8215	0.310	-0.027	3.182
590	0.236	1.167	14	8489	0.314	-0.059	1.895
650	0.235	0.667	14	4264	0.318	-0.078	1.789

Max = **0.425 mm** (LC 15) | L = Member 647 length (1000 mm) | Allowable L/200 = 5.0 mm → **PASS**

CONCLUSION

The structural analysis and design of the **Independent Return** scaffold confirms the structure is adequate for its intended purpose. All member utilization ratios are below unity (maximum UC = **0.352** at Member 1500). Vertical and horizontal deflections are within their respective allowable limits of **L/100 = 15.0mm** and **L/200 = 5.0mm**. All scaffold tube coupler connections are adequate for the applied axial loads.

RECOMMENDATIONS

The scaffold shall be inspected by a competent Scaffold Inspector after erection to confirm conformity with the design specification and approved working drawings before use. All erection work shall comply with the design intent and material specification. Any proposed change to the design requires a re-design and the Scaffold Engineer shall be duly informed.

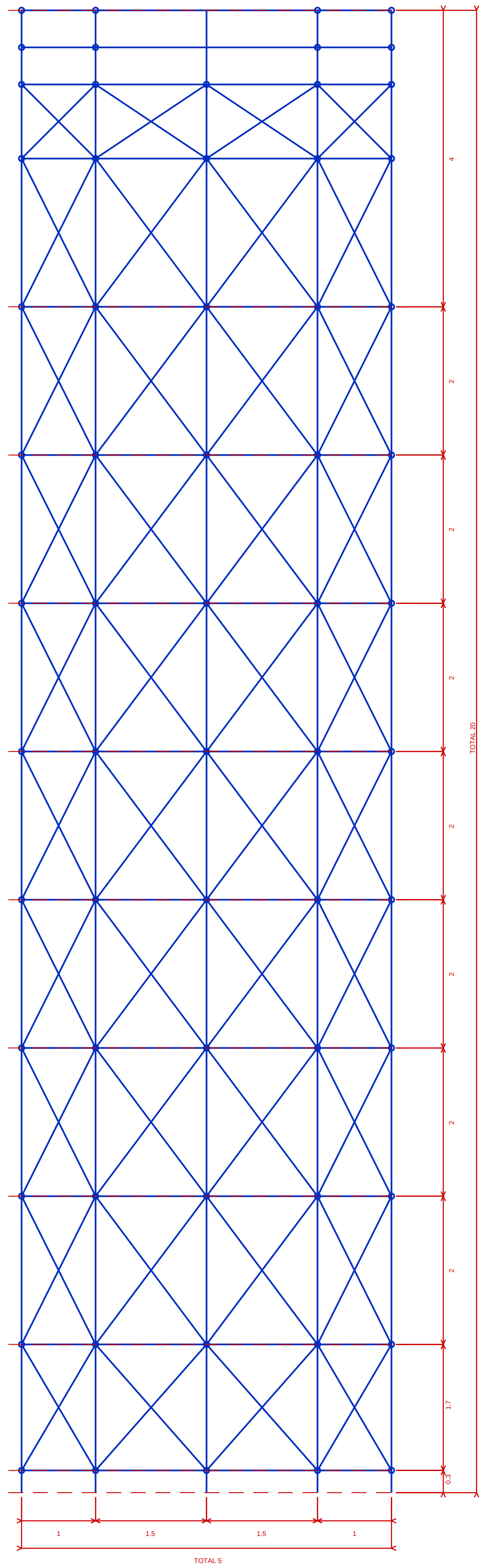
INSTALLATION NOTES

1. The scaffold working drawings and purpose of the scaffold must be reviewed and discussed by all personnel involved prior to execution.
2. Confirm that the Independent Return scaffold configuration matches the approved drawing and is suitable for maintenance.
3. The work vicinity must be barricaded and appropriate signage placed at all access points.
4. All scaffolders must wear complete PPE in accordance with NLNG work-at-height regulations.
5. A valid work permit must be obtained, discussed, and signed before erection commences.
6. Scaffold erection must be carried out in strict accordance with NASC guidance and site safety procedures.
7. Set out and level the scaffold base plates and sole boards in accordance with the approved working drawing.
8. Confirm the scaffold footprint/envelope is approximately 5.0m x 4.0m x 20.0m or as otherwise dimensioned on the approved drawing.
9. Erect scaffold tube standards (48.3 x 3.6 mm), ledgers, and transoms progressively from the base, checking plumb and level in both axes before proceeding to the next lift.
10. Install the lower horizontal ledgers and transoms at the kicker lift height (0.300m) to form the first stable frame.
11. Install mid-level horizontal ledgers and transoms at 2.0m height in accordance with the working drawing.
12. Install mid-level horizontal ledgers and transoms at 4.0m height in accordance with the working drawing.
13. Install mid-level horizontal ledgers and transoms at 6.0m height in accordance with the working drawing.
14. Install mid-level horizontal ledgers and transoms at 8.0m height in accordance with the working drawing.
15. Install mid-level horizontal ledgers and transoms at 10.0m height in accordance with the working drawing.
16. Install mid-level horizontal ledgers and transoms at 12.0m height in accordance with the working drawing.
17. Install mid-level horizontal ledgers and transoms at 14.0m height in accordance with the working drawing.
18. Install mid-level horizontal ledgers and transoms at 16.0m height in accordance with the working drawing.
19. Install mid-level horizontal ledgers and transoms at 18.0m height in accordance with the working drawing.
20. Install mid-level horizontal ledgers and transoms at 19.0m height in accordance with the working drawing.
21. Install mid-level horizontal ledgers and transoms at 19.5m height in accordance with the working drawing.
22. Install all plan, face, and longitudinal bracing shown on the approved drawing to provide sway resistance in both principal directions.
23. Install platform boards, guardrails, mid-rails, toe boards, access arrangements, and any handrail load-resisting members required by the design.
24. Tighten and torque-check all load-bearing couplers and support connections before the scaffold is released for inspection.
25. Gravlock or box-tie the scaffold structure to the existing structure at all indicated tie points at 6.0m and 12.0m height as shown on the working drawing.
26. The scaffold structure must be inspected and tagged by a competent advance scaffold inspector after erection and before use.
27. The safe working load (SWL) stated on the design report must not be exceeded at any time during operations.
28. All modifications made after erection must be carried out by a competent scaffold team with the approval of the scaffold engineer. End-user or third-party modification of this scaffold structure is strictly prohibited.
29. For dismantling, perform the erection steps in reverse order.

REFERENCES & STANDARDS

Reference	Title	Year
BS EN 12811-1:2003	Temporary Works Equipment - Scaffolds - Performance Requirements and General Design	2003
BS EN 1993-1-1:2005	Eurocode 3: Design of Steel Structures - General Rules and Rules for Buildings	2005
BS EN 1991-1-4:2005	Eurocode 1: Actions on Structures - Wind Actions	2005
BS EN 1990:2002	Eurocode: Basis of Structural Design — Load Combinations, Partial Factors, and Limit States (ULS/SLS/EQU)	2002
NASC TG20:13	Guide to Good Practice for Scaffolding with Tubes and Fittings	2013
Durst, C.S.	Wind Speeds Over Short Periods of Time. The Meteorological Magazine, 89, 181–186 — basis of 3-second gust to 10-minute mean conversion	1960
STAAD.Pro CONNECT	STAAD.Pro CONNECT Edition V22 - Structural Analysis and Design Software	2022

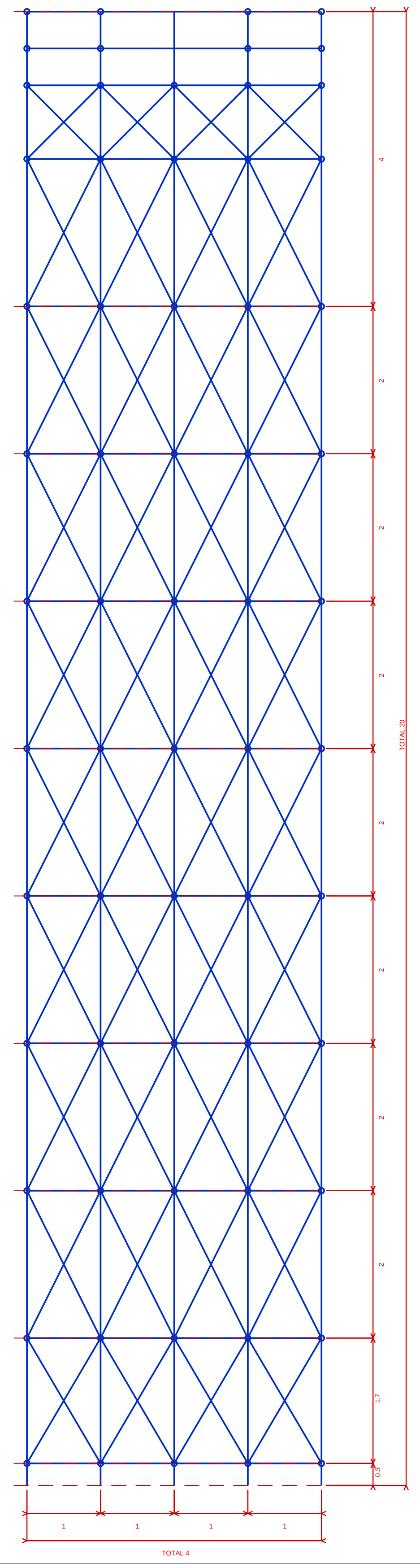
Software Used: STAAD.Pro CONNECT Edition V22 (Bentley Systems) — 3D Structural Analysis and Code Checking per EN 1993-1-1:2005.
Report: Scaffold Structural Design Report — ARCO M&E, NLNG Bonny Island.



FRONT VIEW

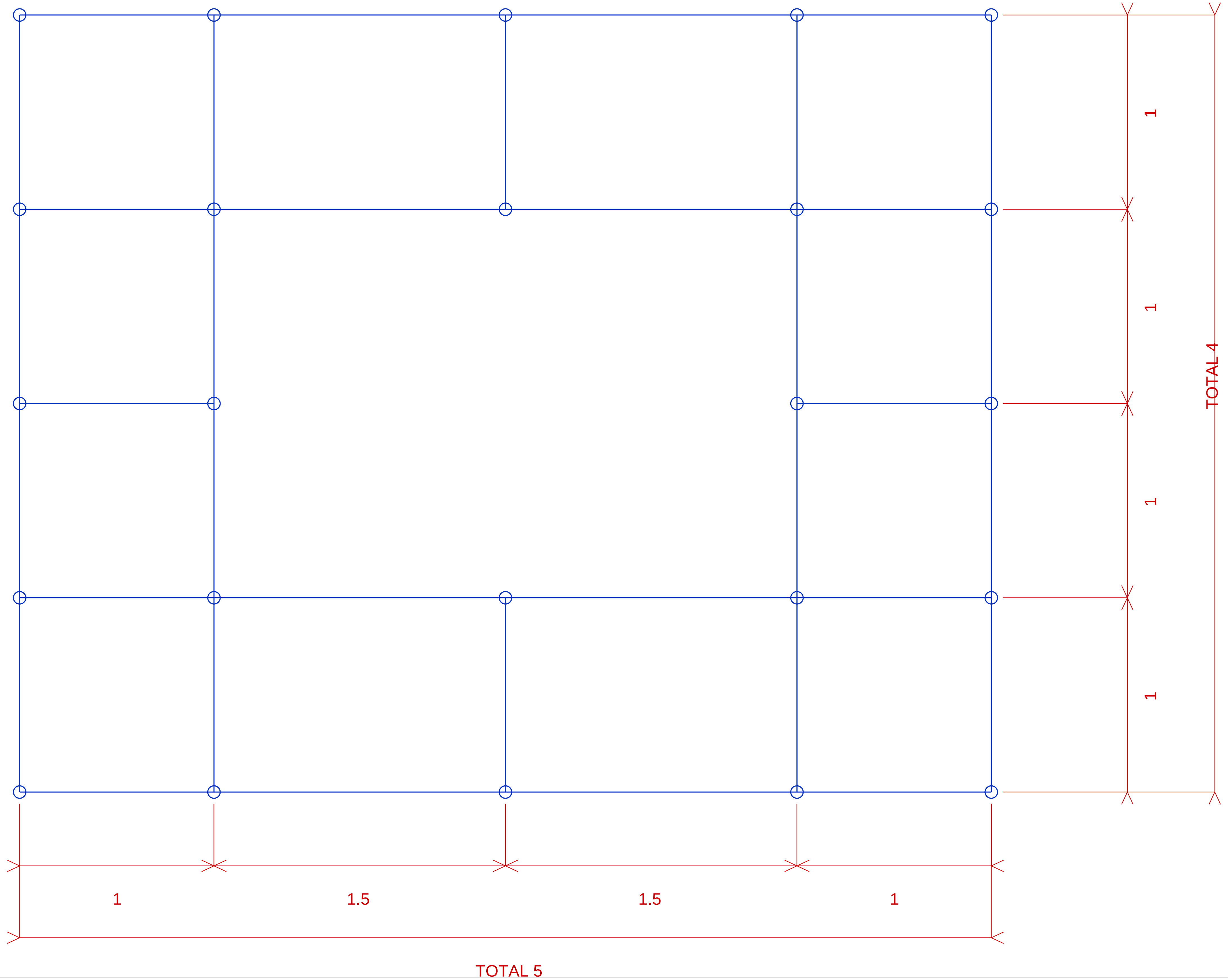
PROJECT **SCAFFOLDING WORKS**		DRAWING NO. **PMS-WA-ARCO-038-FRONT-VIEW**		
DRAWING TITLE **PMS-WA-ARCO-038 - FRONT VIEW**		SCALE **NTS**	DATE **15-JUL-2026**	REV **0**
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	REVIEWED Ezewoko, Joseph	APPROVED Chima, Tasieobi

COMPANY	
SHEET **1 OF 4**	SIZE **A3**



SIDE VIEW

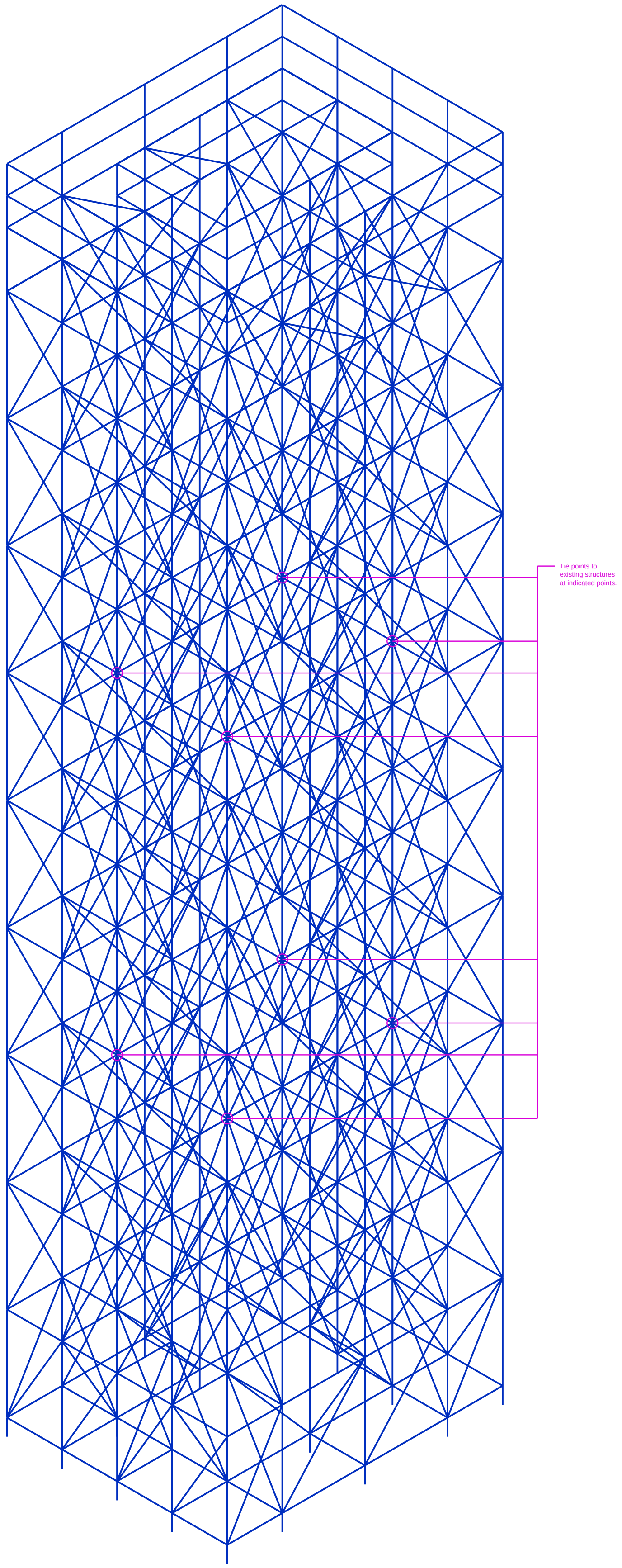
PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-038-SIDE-VIEW			
DRAWING TITLE PMS-WA-ARCO-038 - SIDE VIEW		SCALE NTS	DATE 15-JUL-2026	REV 0	COMPANY
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	REVIEWED Ezewoko, Joseph	APPROVED Chima, Tasieobi	SHEET 2 OF 4 · SIZE A3



PLAN VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-038-PLAN-VIEW		
DRAWING TITLE PMS-WA-ARCO-038 - PLAN VIEW		SCALE NTS	DATE 15-JUL-2026	REV 0
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	REVIEWED Ezewoko, Joseph	APPROVED Chima, Tasieobi

COMPANY	
SHEET 3 OF 4	SIZE A3



3D VIEW

PROJECT **SCAFFOLDING WORKS**		DRAWING NO. **PMS-WA-ARCO-038-3D-VIEW**		
DRAWING TITLE **PMS-WA-ARCO-038 - 3D VIEW**		SCALE **NTS**	DATE **15-JUL-2026**	REV **0**
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	REVIEWED Ezewoko, Joseph	APPROVED Chima, Tasieobi

NLNG

ARCO M&E

COMPANY

SHEET	SIZE
4 OF 4	A3